



SAFETY QUIZ: CHE381- Chemical Engineering Laboratory I

Please answer the following questions testing your knowledge of the content of the Chemical Hygiene Plan safety document. Be brief but concise in your answers. Open book/open notes and team discussion is allowed but answers must be in your own words.

1. What is the most fundamental rule of chemical safety?

Awareness is the most fundamental part of chemical safety

2. What two pieces of personal protective equipment must be worn at all times in the lab?

Safety glasses and safety shoes



- What additional kinds of personal protective equipment are available depending on the chemical hazard or experimental conditions?

Side shields and face masks

4. True or False: Fume hoods are generally used as storage areas for chemicals.

FALSE

5. Fume hoods are to be used when the Threshold Limit Value (TLV) of a chemical is at or below:

1000 ppm or 1000mg/m³

6. What are the four routes of chemical entry into the body?

Inhalation, ingestion, injection, and eye or skin contact

7. True or False: Respirators are generally used in the R&D laboratories for normal work procedures.

FALSE

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8.  What is the role of the Chemical Hygiene Plan?

The plan was designed as a tool to coordinate safety procedures

9. What is the guideline for laboratory room airflow and fume hood airflow?

Laboratory room ventilation should be not less than 20 linear feet per minute of airflow and fume hoods should provide 80-120 linear feet per minute of airflow

10. List 3 Common Housekeeping Practices that promote chemical hygiene and safety:

Keep all aisles, hallways, and stairs clear of all chemicals, all working surfaces and floors should be regularly cleaned, wastes should be properly labeled and kept in the proper containers

 11. What is the proper method of transporting gas cylinders in the laboratory?

Ensure the valve cap is secured and the cylinder is disconnected from regulators. Place and secure the cylinder on a cart using straps or chains, and transport using a cargo elevator.

 12. True or False: MSDS sheets are required only for toxic chemicals and gases used in the laboratory.

TRUE

13. What is an accepted method of vapor detection in the laboratory?

Using mechanical vapor detectors such as Draeger tubes or detection meters is an appropriate way to detect vapor

14. Name 3 Personal Hygiene Practices that should be followed in the laboratory.

Wear appropriate eye protection at all times, do not mouth pipette anything, do not eat or drink in chemical areas

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15. Describe how individual chemicals of less than 15 lbs. (for example a bottle of acetone) are safely transported.

These types of chemicals can be transported using a bucket type carrier that is shock resistant

16. List the Emergency Procedure to be followed for a large chemical spill or toxic gas release in the laboratory

Evacuate the immediate area, dial 0 and advise the nature of emergency as well as location of emergency, exit the lab and deactivate lab power by pushing the red emergency button outside the lab, evacuate the building through the nearest and safest exit, assemble in front of the facility in the middle driveway unless otherwise notified, report to the department secretary

 17. What is the difference between a flammable chemical and a combustible chemical in terms of OSHA and NFPA guidelines.

Flammable chemicals burn at normal working temperature while combustible chemicals need heat to ignite

 18. List the Emergency Procedure for an individual chemical exposure in the laboratory.

Dial 0, advise nature of emergency and location of emergency. If a person has been splashed with a chemical, wash them with plenty of water for at least 15 minutes and remove all contaminated clothing

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19. What is the definition of a “hazardous chemical”?

A chemical is hazardous if it can cause severe health effects such as poisoning, breathing problems, skin rashes and allergic reactions

20. All laboratories that handle chemicals must be equipped with eyewashes and safety showers. What is the recommended flowrate of this equipment and how often should they be checked?

Eyewashes should have a flow rate of 3 gallons per minute and showers should have a flow rate of 30 gallons per minute. Both showers and eyewashes should be checked once a month